Chris Dollo

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Education

University of Maryland Baltimore County Catonsville, MD
Bachelor of Science in Computer Science *Jan 2022 – Dec 2025*

- Relevant Coursework: Data Structures, Software Engineering, Machine Learning, Data Science, Mobile Development, Artificial Intelligence, Computer Security

Experience

Machine Learning Intern May 2025 – Present
Verizon *Catonsville, MD*

- Designed and deployed a scalable end-to-end ML pipeline using YOLOv8 for real-time detection of birds and nests on cell towers, achieving 94% mAP across 14,000+ images.
- Enhanced model robustness through dataset augmentation (rotation, scaling, brightness/contrast, saturation) from Roboflow and Hugging Face sources.
- Improved inference speed and accuracy by optimizing model quantization and preprocessing, reducing false detections by 20%.
- Developed automated testing and validation tools to ensure consistent model reliability within production workflows.
- Collaborated with cross-functional engineering teams to integrate solutions into scalable backend systems for live network inspections.

Undergraduate Research Assistant Jan 2024 – Present
UMBC Vinjamuri Lab *Catonsville, MD*

- Used Python to develop a computer vision pipeline to detect 60 unique hand gestures using MediaPipe.
- Built machine learning and deep learning models using PyTorch to classify EMG signals into 7 hand gesture categories, reaching 87% accuracy.
- Gained hands-on experience with ML model design, debugging, and optimization

Projects

Hellacious | *React.js, Flask, MongoDB, Firebase, AWS S3* Oct 2025 – Present

- Built a scalable full-stack social media platform enabling real-time posting, commenting, and liking using React.js and Flask REST APIs.
- Integrated Firebase Authentication for secure user login and AWS S3 for efficient image storage and retrieval.
- Optimized API endpoints for high-performance media streaming and seamless UI responsiveness.
- Implemented unit tests and version control via Git to maintain reliability in collaborative development.

EcoRecyclr | *PyTorch, Kotlin* Oct 2025 – Nov 2025

- Led a 4-member team to 1st place in the Booz Allen Hamilton Code for Good Hackathon, developing an Android app with on-device ML for real-time waste classification.
- Trained and fine-tuned ResNet-18 on 16,000+ labeled images, achieving 95% classification accuracy.
- Deployed model inference through background services, ensuring non-blocking UI performance and energy-efficient mobile processing.
- Showcased leadership, end-to-end product design, and expertise in scalable mobile ML systems.

Technical Skills

Languages: C, C++, C#, Python, Java, Kotlin, SQL, JavaScript, TypeScript, HTML/CSS, R,
Tools: Git, GitHub, VS Code, PyCharm
Libraries: Pandas, NumPy, PyTorch, TensorFlow, scikit-learn, OpenCV, Seaborn, Matplotlib

Publications

- Olikkal, P., Dollo, C., Ajendla, A. *et al.* Reconstructing hand gestures with synergies extracted from dance movements. *Sci Rep* 15, 41670 (2025). https://doi.org/10.1038/s41598-025-25563-7